

Sri Chandrasekharendra Saraswathi Viswa Mahavidyalaya

[SCSVMV]

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Enathur, Kanchipuram-631561

Department of Computer Science and Engineering



SUBJECT: NATURAL LANGUAGE PROCESSING LAB

CLASS:III BE CSE (AI/ML) V SEM

SECTION: S2 AND S3

ACADEMI YEAR : 2026-2027(JULY-DEC)

FACULTY : Dr.R.PREMA ASSISTANT PROFESSOR

Date: 24.07.2026 Lab hours: 5th and 6th 7th

Laboratory Exercises

Natural Language Processing Lab

Topics

1. Grammar-Based Language Model
2. Statistical Language Model
3. Regular Expressions
4. Finite State Automata (FSA)

These exercises are designed using **Outcome-Based Education (OBE)** and **experiential learning** approaches. Each experiment includes concept understanding, coding, debugging, real-world application, and reflection.

Experiment 1: Grammar-Based Language Model

Learning Outcome

- Understand Context-Free Grammar (CFG).
- Generate valid English sentences using grammar rules.

Activity 1: Predict the Sentence

Given the grammar:

```
S → NP VP
NP → John | Mary
VP → eats NP
```

Students predict all possible sentences before executing the program.

Activity 2: Python Programming

Develop a Python program to generate sentences from the grammar.

Activity 3: Modify the Grammar

Add

NP $\rightarrow$ Student
VP $\rightarrow$ studies NLP

Generate new sentences.

Activity 4: Challenge

Design grammar for

- College
 - Hospital
 - Library
 - Banking System
-

Activity 5: AI Prompt

Prompt ChatGPT

Generate CFG for an Online Shopping System.

Students compare AI-generated grammar with their own.

Mini Project

Design a **Simple English Sentence Generator** using CFG.

Experiment 2: Statistical Language Model

Learning Outcome

- Calculate probabilities using Unigram and Bigram models.
-

Activity 1

Given

```
I like NLP
I like Python
I study NLP
```

Count

- Word frequency
 - Bigram frequency
-

Activity 2

Write a Python program to calculate

- Unigram probability
 - Bigram probability
-

Activity 3

Predict

Which sentence is more probable?

```
I like NLP
Python like I
```

Explain.

Activity 4

Improve Accuracy

Apply Laplace Smoothing.

Observe the difference.

Activity 5

Compare

Grammar-Based Model

vs

Statistical Model

using the same dataset.

Mini Project

Develop a **Next Word Prediction System**.

Example

Input

I love

Output

Python

NLP

AI

Experiment 3: Regular Expressions

Learning Outcome

Extract useful information from text.

Activity 1

Extract

- Email IDs
- Mobile Numbers
- URLs

from a paragraph.

Activity 2

Complete the Regex

_____@gmail.com

Students write the missing expression.

Activity 3

Debugging

Given

[a-z

Find the error.

Activity 4

Real-world Exercise

Given a Resume

Extract

- Name
 - Email
 - Phone Number
 - Skills
-

Activity 5

Pattern Competition

Groups write regex for

- Aadhaar Number
- PAN Number
- Vehicle Number
- Date
- Time

Most efficient regex wins.

Mini Project

Build a **Resume Information Extractor**.

Experiment 4: Finite State Automata

Learning Outcome

Design DFA and implement it in Python.

Activity 1

Draw DFA accepting

ab^*

Students draw manually.

Activity 2

Python Implementation

Implement DFA.

Test

a
ab
abb
abbb

Activity 3

Debugging

Provide incorrect transition table.

Students correct it.

Activity 4

Simulation Game

Assign students

State q0

State q1

State q2

One student acts as input symbols.

Others simulate DFA.

Activity 5

Design Challenge

Design DFA for

- Binary numbers divisible by 2
 - Valid identifier
 - Password validation
 - Vowel recognition
-

Mini Project

Develop a **DFA Simulator**.

Integrated Lab Challenge

Combine all four concepts.

Problem Statement

Develop a **Mini NLP Text Analyzer**

Features

- ✓ Read text
- ✓ Clean text using Regular Expressions
- ✓ Validate words using DFA

- ✓ Calculate Bigram Probability
 - ✓ Generate sentence using Grammar
-

Gamified Lab Activities

Activity	Marks
Predict Output	2
Coding	4
Debugging	2
Mini Challenge	1
Reflection	1

Total = **10 Marks**

Think-Pair-Share Questions

- Why is Grammar-Based LM less flexible?
 - Why is Statistical LM more accurate?
 - Where are Regular Expressions used in industry?
 - Why is DFA important in NLP?
 - Can Regex replace Grammar? Why?
-

Real-World Applications

Topic	Application
Grammar-Based LM	Chatbots, Language Tutors
Statistical LM	Search Engines, Autocomplete
Regular Expressions	Spam Detection, Data Cleaning
Finite State Automata	Lexical Analysis, Tokenization, Compiler Design

Lab Assessment Rubric (10 Marks)

Criteria	Marks
Problem Understanding	2

Criteria	Marks
Algorithm Design	2
Python Program	3
Output Verification	1
Innovation/Extension	1
Viva Voce	1
Total	10

These activities promote **active learning, critical thinking, debugging skills, teamwork, and real-world problem solving**, making the NLP laboratory more engaging while covering Grammar-Based Language Models, Statistical Language Models, Regular Expressions, and Finite State Automata.